

| Register(s)    | Alt.   | Description                                                       |
|----------------|--------|-------------------------------------------------------------------|
| x0             | zero   | The zero register, always zero                                    |
| x1             | ra     | The return address register, stores where functions should return |
| x2             | sp     | The stack pointer, where the stack ends                           |
| x5-x7, x28-x31 | t0-t6  | The temporary registers                                           |
| x8-x9, x18-x27 | s0-s11 | The saved registers                                               |
| x10-x17        | a0-a7  | The argument registers, a0-a1 are also return value               |

| Encoding | Register   | Alt. Name | Description                                                   |
|----------|------------|-----------|---------------------------------------------------------------|
| 00000    | <b>x0</b>  | zero      | The Zero Register, always zero                                |
| 00001    | <b>x1</b>  | ra        | The Return Address register, where the function should return |
| 00010    | <b>x2</b>  | sp        | The Stack Pointer, where the stack ends                       |
| 00101    | <b>x5</b>  | t0        | The Temporary Registers                                       |
| 00110    | <b>x6</b>  | t1        |                                                               |
| 00111    | <b>x7</b>  | t2        |                                                               |
| 11100    | <b>x28</b> | t3        |                                                               |
| 11101    | <b>x29</b> | t4        |                                                               |
| 11110    | <b>x30</b> | t5        |                                                               |
| 11111    | <b>x31</b> | t6        |                                                               |
| 01000    | <b>x8</b>  | s0        | The Saved Registers                                           |
| 01001    | <b>x9</b>  | s1        |                                                               |
| 10010    | <b>x18</b> | s2        |                                                               |
| 10011    | <b>x19</b> | s3        |                                                               |
| 10100    | <b>x20</b> | s4        |                                                               |
| 10101    | <b>x21</b> | s5        |                                                               |
| 10110    | <b>x22</b> | s6        |                                                               |
| 10111    | <b>x23</b> | s7        |                                                               |
| 11000    | <b>x24</b> | s8        |                                                               |
| 11001    | <b>x25</b> | s9        |                                                               |
| 11010    | <b>x26</b> | s10       |                                                               |
| 11011    | <b>x27</b> | s11       |                                                               |
| 01010    | <b>x10</b> | a0        | The Arguments Registers, x10-x11 are also return values       |
| 01011    | <b>x11</b> | a1        |                                                               |
| 01100    | <b>x12</b> | a2        |                                                               |
| 01101    | <b>x13</b> | a3        |                                                               |
| 01110    | <b>x14</b> | a4        |                                                               |
| 01111    | <b>x15</b> | a5        |                                                               |
| 10000    | <b>x16</b> | a6        |                                                               |
| 10001    | <b>x17</b> | a7        |                                                               |

*The professor doesn't allow those tables during the exam.*